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Msc Ds DS602

Lab Tutorial

Step 1 : Creating Venv

“python -m venv venv”

A virtual environment creates a separate Python environment specifically for this project.

“pip install -r requirements.txt”

The objective of this step was to install all the Python libraries required for performing data analysis, statistical modeling, visualization, hypothesis testing, and developing the interactive Streamlit dashboard after creating virtual environment venv.

The assignment recommends using Python along with tools such as statsmodels, Streamlit, and VS Code.

Library	Purpose in the Project
Pandas	Used to load, organize, clean, and manipulate tabular data.
NumPy	Used for numerical and mathematical operations.
Matplotlib	Used to create basic data visualizations.
Seaborn	Used to create statistical and exploratory data visualizations.
SciPy	Used to perform statistical hypothesis tests.
Statsmodels	Used to build the OLS regression model and perform statistical analysis.
Scikit-learn	Used to access and work with the California Housing dataset.
Streamlit	Used to develop the interactive web dashboard.
Plotly	Used to create interactive visualizations for the dashboard.

Step 2: Understanding the dataset

- The dataset was successfully loaded into a Pandas DataFrame.
- It contains **20,640 observations and 9 variables**.
- All the original variables are numerical and have the float64 data type.
- The dataset contains **no missing values**.
- **MedHouseVal** will be used as the primary target variable for regression modeling.
- Additional categorical variables will later be created to support group-based hypothesis testing and interactive dashboard filtering.

```
MedInc HouseAge AveRooms AveBedrms Population AveOccup Latitude \
0 8.3252 41.0 6.984127 1.023810 322.0 2.555556 37.88
1 8.3014 21.0 6.238137 0.971880 2401.0 2.109842 37.86
2 7.2574 52.0 8.288136 1.073446 496.0 2.802260 37.85
3 5.6431 52.0 5.817352 1.073059 558.0 2.547945 37.85
4 3.8462 52.0 6.281853 1.081081 565.0 2.181467 37.85

Longitude MedHouseVal
0 -122.23 4.526
1 -122.22 3.585
2 -122.24 3.521
3 -122.25 3.413
4 -122.25 3.422

Dataset Shape:
(20640, 9)

Column Names:
Index(['MedInc', 'HouseAge', 'AveRooms', 'AveBedrms', 'Population', 'AveOccup',
       'Latitude', 'Longitude', 'MedHouseVal'],
      dtype='str')
```

Step 3:

Creation of Categorical columns

AgeGroup and IncomeGroup variables using `pd.cut()` and `pd.qcut()`.

The original California Housing dataset contains only numerical variables. However, categorical variables are useful for comparing groups during hypothesis testing and for creating interactive filters in the dashboard.

Therefore, two new categorical variables were created:

1. **AgeGroup** – based on HouseAge

0–20 years → New

20–40 years → Middle-aged

More than 40 → Old

2. IncomeGroup – based on MedInc

These newly created variables do not replace the original numerical columns. Instead, they provide additional categories that can be used for group-based statistical analysis.

I used the `pd.qcut()` function with `q=3`. This first considers all the income values in the dataset and divides them into **three approximately equal-sized groups**:

- The lowest one-third of income values → **Low**
- The middle one-third → **Medium**
- The highest one-third → **High**

Create Categorical Groups

```
#Age Groups
df["AgeGroup"] = pd.cut(
    df["HouseAge"],
    bins=[0, 20, 40, float("inf")],
    labels=["New", "Middle-aged", "old"]
)

#income groups
df["IncomeGroup"] = pd.qcut(
    df["MedInc"],
    q=3,
    labels=["Low", "Medium", "High"]
)
```

```
df.head()
```

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal	AgeGroup	IncomeGroup
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526	Old	High
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585	Middle-aged	High
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521	Old	High
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25	3.413	Old	High
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25	3.422	Old	Medium

Step 4: Descriptive Statistics

1. Mean

The mean represents the arithmetic average of all observations.

Formula

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

Where:

- x_i = each observation
- n = total number of observations

In project

The mean of MedHouseVal represents the average median house value across all California districts in the dataset.

Median

The median is the middle value when the observations are arranged in ascending order.

It is less affected by extreme values than the mean.

Standard Deviation

Standard deviation measures the amount of variation in the data.

Formula

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

A higher standard deviation indicates greater variability.

IQR

The Interquartile Range measures the spread of the middle 50% of the data.

Formula

$$\text{IQR} = Q3 - Q1$$

Where:

- $Q1$ = 25th percentile
 - $Q3$ = 75th percentile
-

Skewness

Skewness describes the asymmetry of a distribution.

Skewness ≈ 0 → Approximately symmetric

Skewness > 0 → Right-skewed

Skewness < 0 → Left-skewed

Kurtosis

Kurtosis describes the tail heaviness or extremity of a distribution.

It helps us understand whether extreme values may be more or less prominent than in a normal distribution.

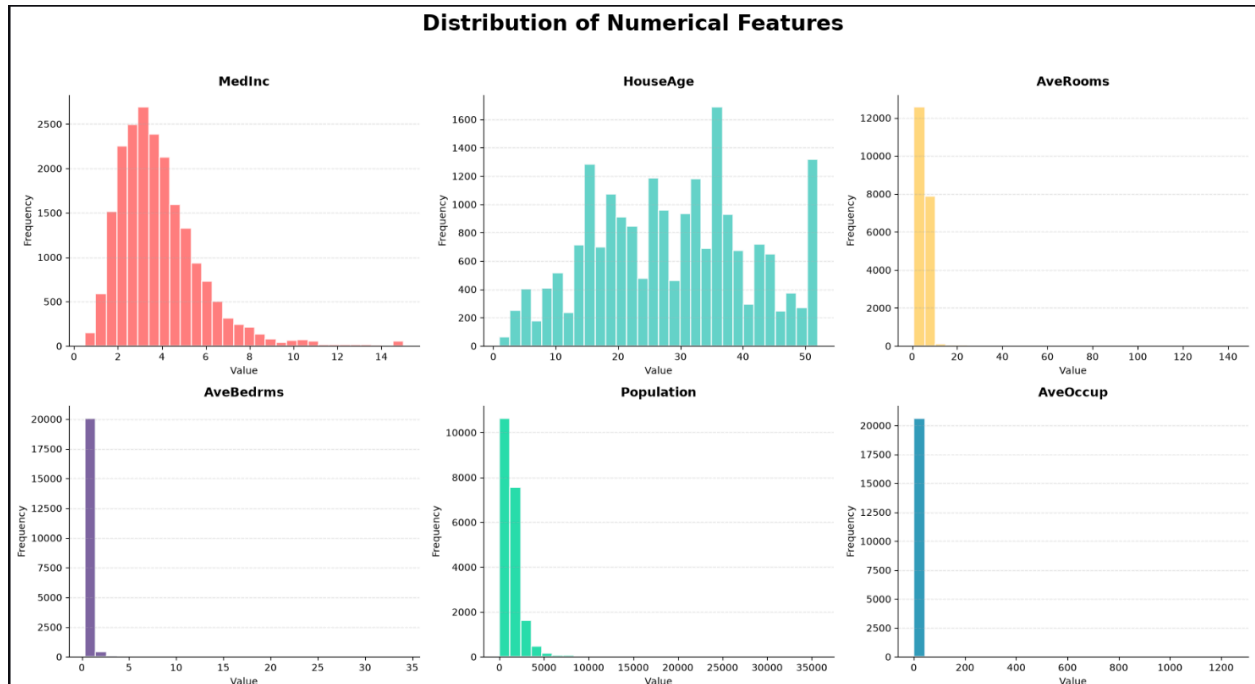
	Mean	Median	Standard Deviation	IQR	\
MedInc	3.870671	3.534800	1.899822	2.179850	
HouseAge	28.639486	29.000000	12.585558	19.000000	
AveRooms	5.429000	5.229129	2.474173	1.611665	
AveBedrms	1.096675	1.048780	0.473911	0.093447	
Population	1425.476744	1166.000000	1132.462122	938.000000	
AveOccup	3.070655	2.818116	10.386050	0.852520	
Latitude	35.631861	34.260000	2.135952	3.780000	
Longitude	-119.569704	-118.490000	2.003532	3.790000	
MedHouseVal	2.068558	1.797000	1.153956	1.451250	

	Skewness	Kurtosis
MedInc	1.646657	4.952524
HouseAge	0.060331	-0.800629
AveRooms	20.697869	879.353264
AveBedrms	31.316956	1636.711972
Population	4.935858	73.553116
AveOccup	97.639561	10651.010636
Latitude	0.465953	-1.117760
Longitude	-0.297801	-1.330152
MedHouseVal	0.977763	0.327870

Step 5: Visual Exploratory Data Analysis

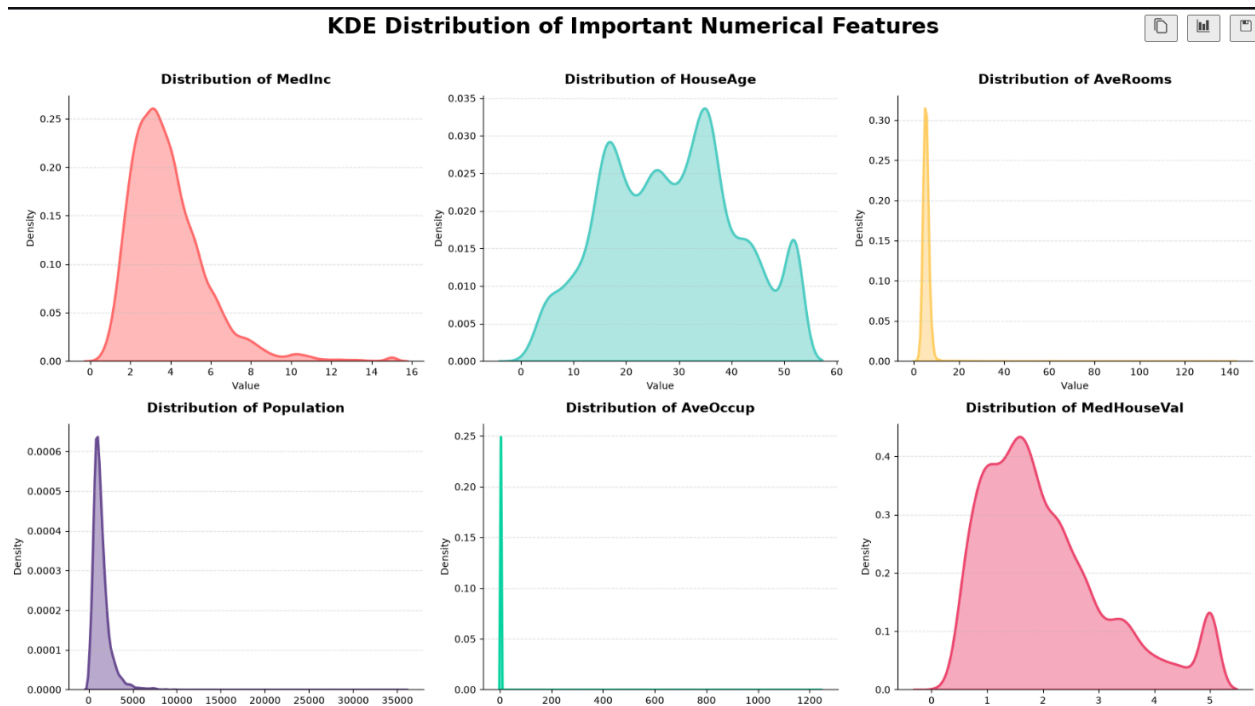
1. Histograms for numerical values:

Generated to examine the distribution of numerical variables.



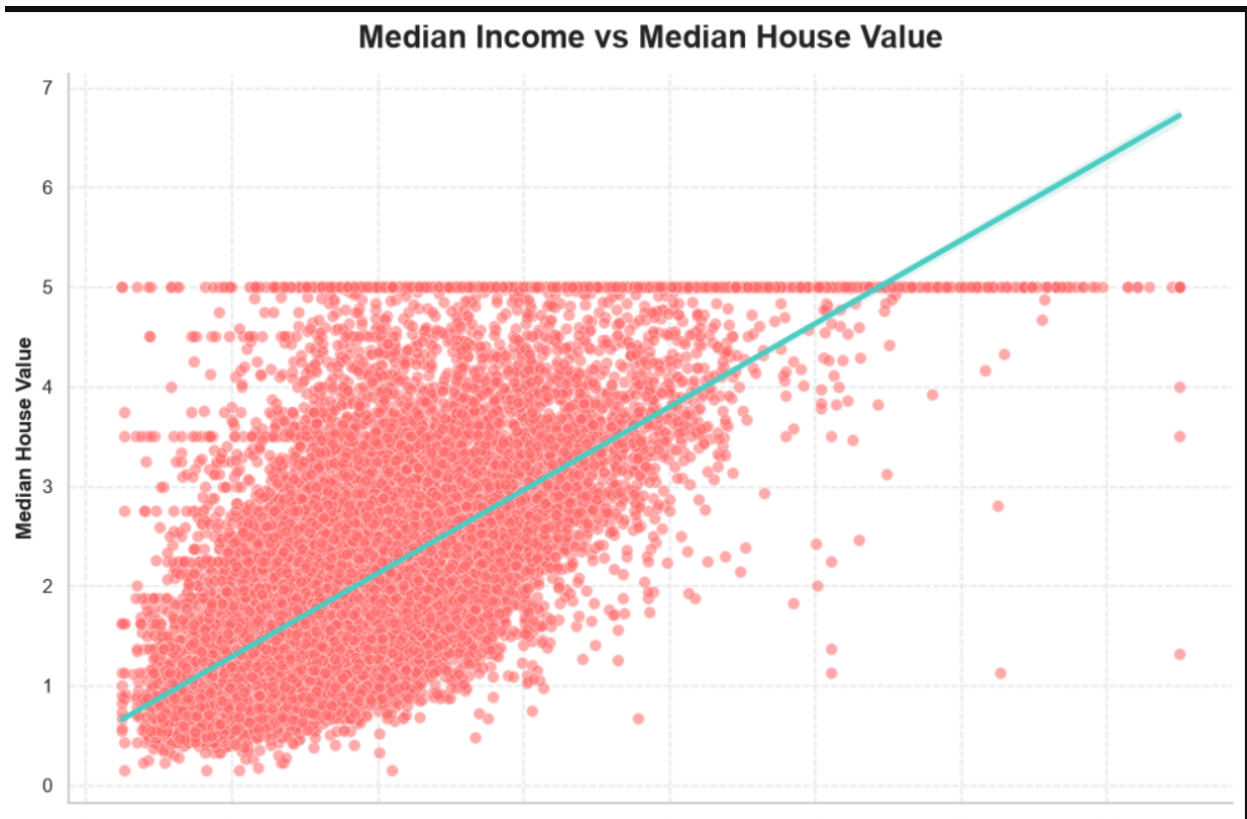
2. Kernel Density Estimation Plots

A smooth, continuous curve that shows the probability distribution of a continuous variables(histogram)



3.Scatter plot:

i) To examine the relationship between median income and median house value.



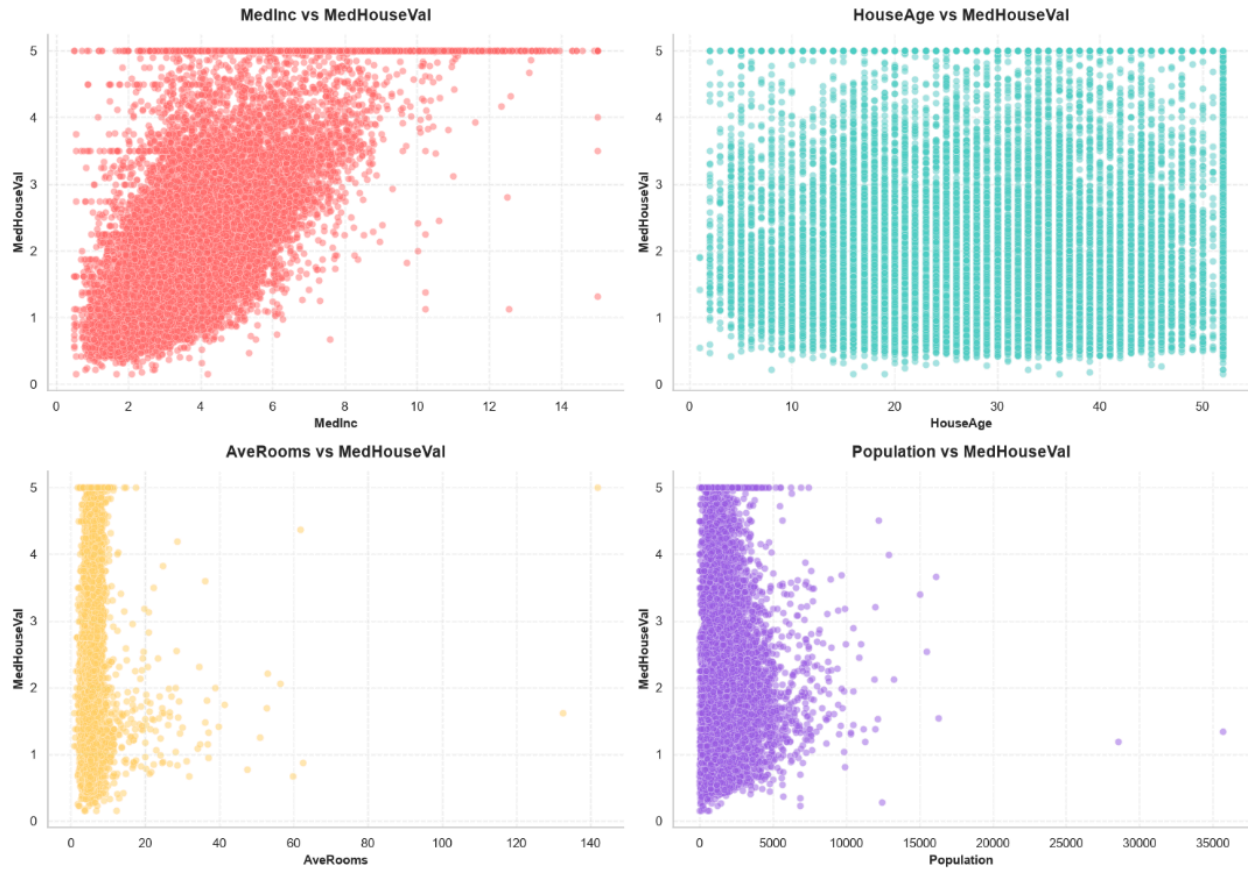
The scatter plot shows a positive association between median income and median house value. In general, areas with higher median income tend to have higher median house values. However, considerable variation exists, indicating that median income alone does not completely explain house values.

Additional Observation: A concentration of observations appears at the maximum house value of 5, indicating a possible upper-value cap in the dataset.

Conclusion: Median income appears to be an important predictor of median house value and will therefore be included in the multiple linear regression analysis.

ii) Multiple relationships with our target variable, *MedHouseVal*

Relationship Between Numerical Features and House Value



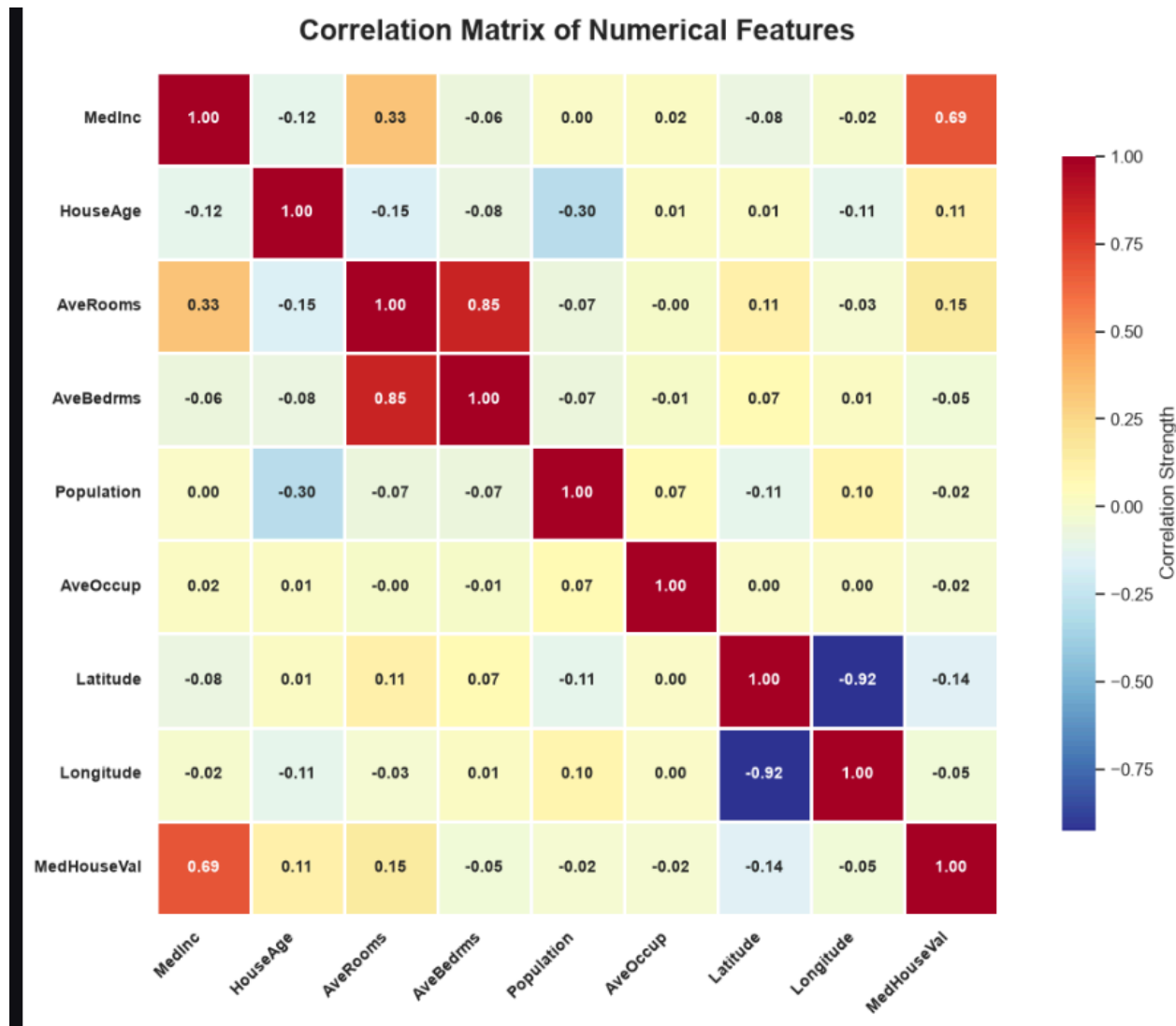
The scatter plots showed that MedInc has the clearest visible positive relationship with MedHouseVal, where areas with higher median income generally tend to have higher house values. The relationship between HouseAge and house value appeared weaker, with substantial variation in house values across different age groups. The AveRooms and Population plots showed that most observations were concentrated within lower ranges, while a small number of observations extended to very high values, supporting the earlier findings of high positive skewness and potential outliers

3. Correlation Heatmap

Visualization of correlation matrix.

A correlation analysis was performed to measure the **strength and direction of the linear relationship** between the numerical variables in the California Housing dataset. Pearson's correlation coefficient was calculated for all numerical features, and the results were visualized using a correlation heatmap.

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$



- The correlation matrix showed that MedInc had the strongest positive linear relationship with the target variable, MedHouseVal, with a correlation coefficient of approximately 0.69.
- This suggests that higher median income is generally associated with higher median house values.

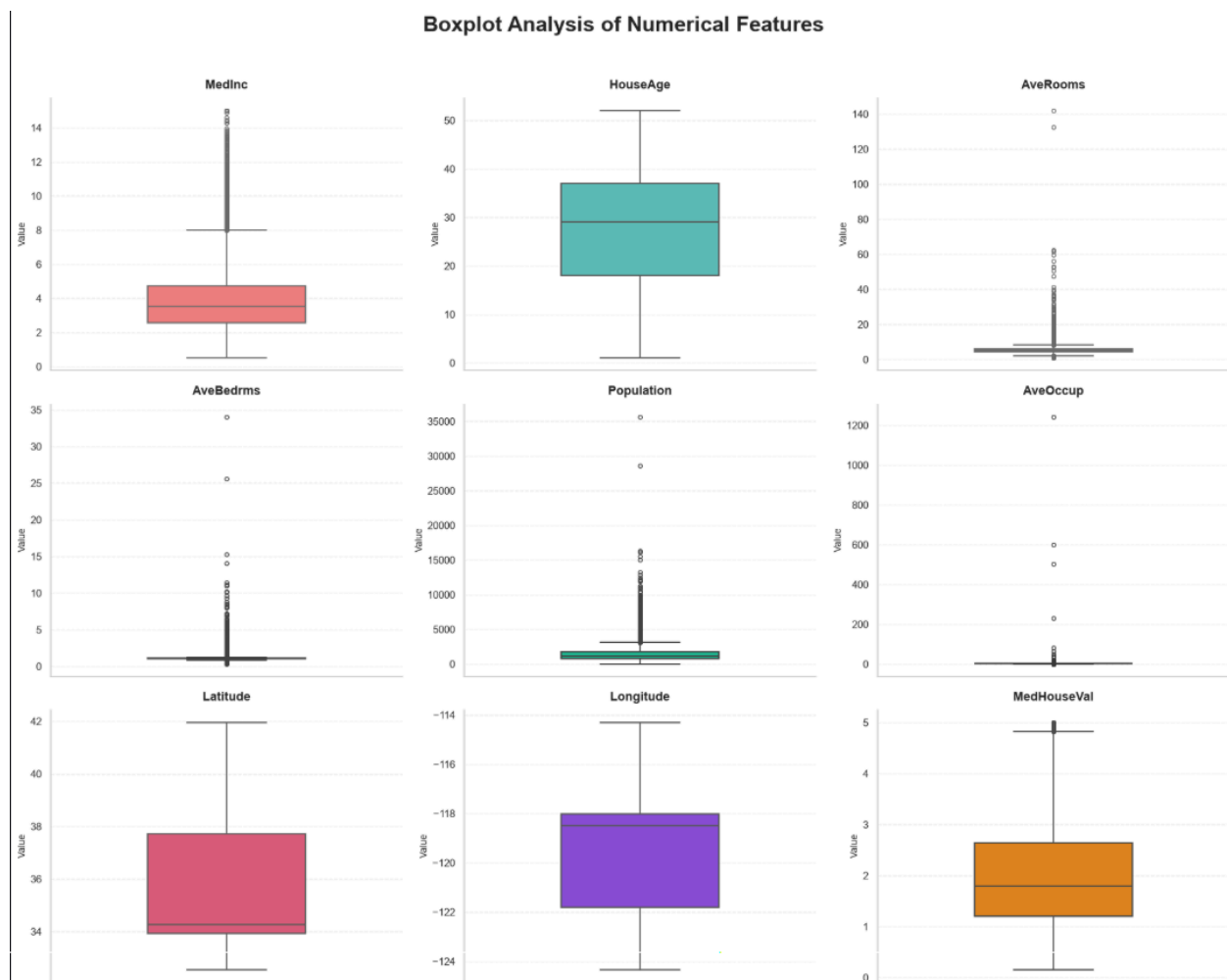
- The analysis also revealed a strong positive correlation between AveRooms and AveBedrms (0.85) and a strong negative correlation between Latitude and Longitude (-0.92).
- These strong relationships between predictor variables may indicate potential multicollinearity, which will be examined later using Variance Inflation Factor (VIF). Most of the remaining variables showed relatively weak linear correlations with MedHouseVal.

4) Boxplot

A boxplot helps us visualize:

- Median
- Quartiles
- IQR
- Potential outliers

$IQR = Q3 - Q1$ (tells far from the middle 50% of the observations.)



- Boxplots were generated to examine the distribution and potential outliers in the numerical variables. HouseAge, Latitude, and Longitude showed relatively limited evidence of extreme observations compared with the other variables.
- In contrast, MedInc, AveRooms, AveBedrms, Population, and particularly AveOccup showed observations beyond their upper whiskers. The presence of extreme values is consistent with the high positive skewness and kurtosis observed in the descriptive statistics.
- AveOccup exhibited the most extreme distribution, with a small number of observations substantially higher than the majority of the data.
- The target variable MedHouseVal also showed a concentration at its upper limit of 5, indicating a ceiling effect.

Variable	Potential Outliers	Interpretation
MedInc	681	Some unusually high-income districts
HouseAge	0	No IQR-based outliers
AveRooms	511	Several extreme values
AveBedrms	1,424	Many potential extreme values
Population	1,196	Many high-population observations
AveOccup	711	Strong presence of extreme values
Latitude	0	No IQR-based outliers
Longitude	0	No IQR-based outliers
MedHouseVal	1,071	Many values above the IQR upper bound

Step 6: Hypothesis testing

Pick two groups and compare a continuous metric.

It specifically says we must:

1. Formulate H_0 and H_1
2. Check normality using **Shapiro-Wilk**
3. Check equal variance using **Levene's test**

4. If normal → **Two-Sample t-test**
5. If not normal → **Mann-Whitney U**
6. Make the decision at $\alpha=0.05$

Hypothesis Test 1: Low Income vs High Income using target numerical variable MedHouseVal .

Using:

- Grouping variable: IncomeGroup
- Group 1: Low
- Group 2: High
- Numerical variable: MedHouseVal

As we know :

Null hypothesis H0 -> H0: The data is normally distributed

Alternative hypothesis H1-> H1: The data is not normally distributed

Normality Test — Shapiro-Wilk

The Shapiro-Wilk test was performed to assess whether the MedHouseVal observations in the low-income and high-income groups followed a normal distribution. The null hypothesis stated that the observations were normally distributed, while the alternative hypothesis stated that they were not normally distributed.

For the low-income group, the Shapiro-Wilk statistic was 0.8491 with a p-value of 2.20×10^{-56} . Since the p-value was less than the significance level of 0.05, the null hypothesis was rejected, indicating evidence of non-normality.

For the high-income group, the Shapiro-Wilk statistic was 0.9340 with a p-value of 1.78×10^{-42} . Since this p-value was also less than 0.05, the null hypothesis was rejected, indicating evidence of non-normality.

Therefore, the normality assumption was not supported for either group. Following the decision procedure specified in the assignment, a non-parametric Mann-Whitney U test was selected for comparing the two groups.

We used the Shapiro-Wilk test, because the Shapiro-Wilk test indicated that the house-value distributions in both groups were not normally distributed. According to the decision rule given in the lab, we therefore use the non-parametric Mann-Whitney U test.

Check normality $\rightarrow p < 0.05 \rightarrow$ Reject normality $\rightarrow$ Mann-Whitney U

Step 7: One-Way ANOVA

For comparing more than two groups, **One-Way ANOVA** was performed.

- Grouping variable: AgeGroup
- Groups: New, Middle-aged, Old
- Target variable: MedHouseVal
- Significance level: $\alpha = 0.05$

Hypotheses

H₀: The mean house values are equal across all age groups.

H₁: At least one age group has a different mean house value.

The ANOVA test was performed using `scipy.stats.f_oneway()`.

If the ANOVA p-value is less than 0.05, H₀ is rejected. When significant, **Tukey HSD** is used to identify which groups differ.

Result: The ANOVA result is displayed interactively in the Streamlit dashboard.

Step 8: Multiple Linear Regression

A **Multiple Linear Regression model using OLS** was developed to predict MedHouseVal.

Predictor Variables

- MedInc
- HouseAge
- AveRooms
- AveBedrms
- Population
- AveOccup

- Latitude
- Longitude

The model was implemented using **Statsmodels OLS**.

Regression Equation

$$MedHouseVal = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_8 X_8 + \epsilon$$

The dashboard displays:

- Regression coefficients
 - p-values
 - 95% confidence intervals
 - R^2
 - Adjusted R^2
-

Step 9: Regression Diagnostics

Several diagnostic methods were used to evaluate the OLS model.

Residual vs Fitted Plot

Used to check whether residuals are randomly distributed around zero.

Q-Q Plot

Used to check whether residuals approximately follow a normal distribution.

Jarque-Bera Test

Used to statistically test residual normality.

- H_0 : Residuals are normally distributed.
- H_1 : Residuals are not normally distributed.

VIF

Variance Inflation Factor was used to check **multicollinearity** among predictor variables.

Step 10: Live Prediction

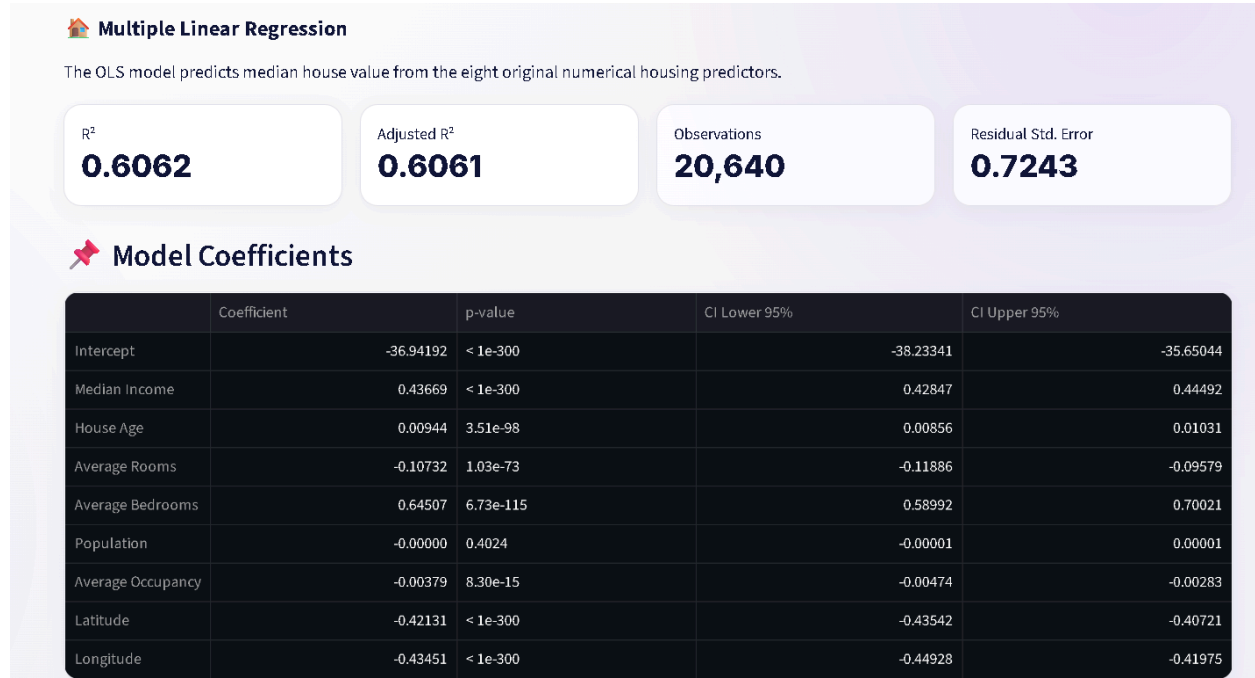
The dashboard provides a live prediction feature where users can enter values for the eight predictor variables.

The OLS model generates the predicted median house value.

The dashboard displays:

- **Predicted House Value**
- **95% Confidence Interval**
- **95% Prediction Interval**

The original MedHouseVal values are in units of \$100,000, so the dashboard converts the prediction into dollar values.



Step 11: Streamlit Dashboard

The complete analysis was converted into an interactive Streamlit application.

The dashboard contains three tabs:



Data Exploration

- Dataset overview
- Descriptive statistics
- Histograms
- Boxplots
- Scatter plots
- Correlation matrix



Hypothesis Testing Lab

- Shapiro-Wilk
- Levene's test
- t-test / Mann-Whitney U
- One-Way ANOVA
- Tukey HSD



Live Prediction & Diagnostics

- OLS regression
 - Prediction
 - Confidence and prediction intervals
 - Residual diagnostics
 - Jarque-Bera
 - VIF
-

Step 12: Deployment

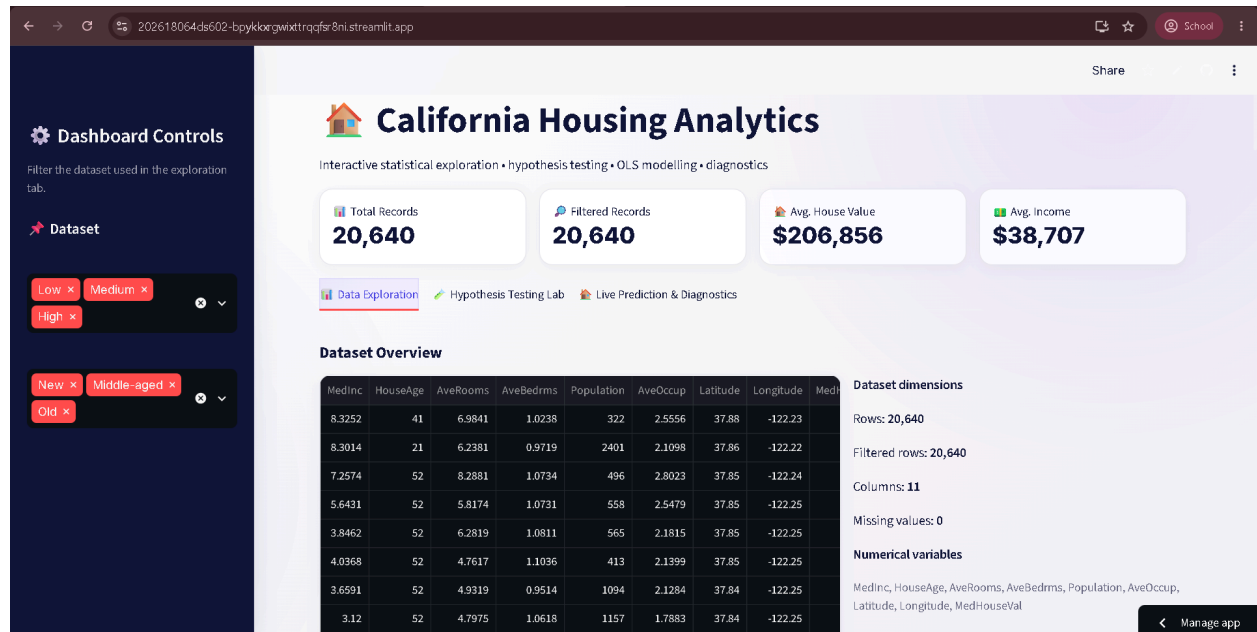
The Streamlit application was deployed using **Streamlit Community Cloud**.

GitHub Repository

https://github.com/202618064-dau/202618064_DS602

Live Dashboard

<https://202618064ds602-bpykkxrgvvixtqqfsr8ni.streamlit.app>



Conclusion

The California Housing dataset was analyzed using descriptive statistics, visualization, hypothesis testing, and OLS regression.

The analysis showed a strong positive relationship between **median income (MedInc)** and **median house value (MedHouseVal)**. The Low Income and High Income groups also showed a statistically significant difference in house-value distributions.

Finally, all statistical analyses were integrated into an interactive **Streamlit dashboard** with live prediction and regression diagnostics.
